

Assignment 2: Clustering and Dimensionality Reduction

1 Introduction

In this assignment, you will apply and compare dimensionality reduction and clustering techniques on real multidimensional datasets.

The assignment can be submitted as a Jupyter notebook containing code, text, and figures, or as source code together with a PDF report.

Select three datasets from the following list:

- [Dry Bean](#)
- [MAGIC Gamma Telescope](#)
- [Cardiotocography](#)
- [Pen-Based Recognition of Handwritten Digits](#)
- [Optical Recognition of Handwritten Digits](#)

For each dataset, briefly describe the data and any preprocessing performed.

2 Dimensionality Reduction

Apply the following techniques to each dataset:

- Principal Component Analysis (PCA)
- t-SNE
- UMAP

Reduce each dataset to two dimensions and visualize the results using scatterplots. Color the observations according to their class labels.

Grade G

For Grade G:

- Apply all three techniques correctly.
- Present and compare the resulting visualizations.
- Discuss which technique provides the clearest class separation for each dataset.
- Discuss differences in the structures revealed by the three techniques.

Grade VG

In addition to the requirements for Grade G:

- Investigate important parameter settings for t-SNE and UMAP and discuss how they affect the results.
- Use at least one quantitative measure of dimensionality reduction quality, such as trustworthiness or neighbourhood preservation.
- Combine the quantitative and visual results to determine which technique is most suitable for each dataset.

3 Clustering

Apply the following techniques to each dataset:

- K-Means
- DBSCAN
- Hierarchical Clustering

Apply clustering to the original multidimensional data. Use one of the dimensionality reduction techniques from the previous section only to visualize the clustering results in two dimensions.

Grade G

For Grade G:

- Apply all three clustering techniques correctly.
- Visualize and compare the resulting clusters.
- Discuss which technique produces the most meaningful clustering for each dataset.
- Compare the identified clusters with the original class labels.

Grade VG

In addition to the requirements for Grade G:

- Investigate important parameter settings for at least two clustering techniques and discuss their influence on the results.
- Evaluate the clustering using at least one internal measure, such as the Silhouette Score.
- Compare the clusters with the original class labels using at least one external measure, such as Adjusted Rand Index.
- Use the quantitative and visual results to justify which clustering technique is most suitable for each dataset.

4 Grading

Grade G requires satisfactory completion of all Grade G tasks for both dimensionality reduction and clustering.

Grade VG requires satisfactory completion of all Grade G tasks and all additional Grade VG tasks.